Andover

$$\begin{aligned}\frac{E_A}{K} &= 1 \cdot \binom{4}{1} pq^3 + 4\left(1 - q^4 - \binom{4}{1} pq^3\right) \\ &= 4pq^3 + 4(1 - q^4 - 4pq^3) \\ &= 4p(1 + q + q^2 - 2q^3)\end{aligned}$$

Basingstoke

$$\begin{aligned}\frac{E_B}{K} &= 1 \cdot \binom{6}{1} pq^5 + 2 \cdot \binom{6}{2} p^2q^4 + 6\left(1 - q^6 - \binom{6}{1} pq^5 - \binom{6}{2} p^2q^4\right) \\ &= 6pq^5 + 30p^2q^4 + 6(1 - q^6 - 6pq^5 - 15p^2q^4) \\ &= 6 - 60q^4 + 90q^5 - 36q^6 \\ &= 6(1 - q)(1 + q + q^2 + q^3 - 9q^4 + 6q^5)\end{aligned}$$

$$E_B = 6Kp(1 + q + q^2 + q^3 - 9q^4 + 6q^5)$$

$$E_A = \frac{2}{3} E_B$$

$$4Kp(1 + q + q^2 - 2q^3) = \frac{2}{3} \cdot 6Kp(1 + q + q^2 + q^3 - 9q^4 + 6q^5)$$

For $p = 0$,

$$E_A = E_B = 0$$

For $p > 0$,

$$1 + q + q^2 - 2q^3 = 1 + q + q^2 + q^3 - 9q^4 + 6q^5$$

$$3q^3(1 - q)(1 - 2q) = 0$$

$$\implies p = 1,\ \frac{1}{2}$$

This solution was submitted by **Jon**
Published and refined by **Jon**

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